

Evaluation, Deployment and Monitoring of Models

Task: XAI3

Model-Agnostic Methods: Partial Dependence Plots

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Subject

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1 Introduction

This report solves Exercise 5 on model-agnostic Partial Dependence Plots (PDP). The objective is to fit random forest regression models and use PDPs to interpret how selected input variables affect the predicted target.

For the bike rental problem, the target variable is `cnt`. The daily bike dataset was used because the question asks for the influence of days since 2011. The model uses calendar, weather, and day-type variables, including `days_since_2011`, `temp`, `hum`, and `windspeed`.

For the house price problem, the target variable is `price`. The model uses `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, and `yr_built`, as requested in the exercise.

All random choices use a fixed seed of 42. The PDPs are generated with `scikit-learn`. Random samples are used for the larger house dataset and for the bidimensional PDP step, following the warning in the exercise statement.

2 Data Preparation and Models

The bike-sharing data contain daily rental counts and weather/calendar variables. The original date field is transformed into `days_since_2011`, which is the variable requested in the statement. The target is the total number of daily rentals, `cnt`. The variables `casual` and `registered` are excluded because they sum directly to `cnt` and would leak the answer.

The house-price data are modelled with the variables requested in the task: `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, and `yr_built`. The target is `price`. Random forest regressors are used in both exercises because they provide flexible non-linear approximations while remaining compatible with model-agnostic PDP analysis.

Model	Test R^2	Test MAE
Bike rentals	0.873	450.43 rentals
House prices	0.610	156,514.67 USD

Table 1: Validation metrics for the random forest regressors.

3 One-Dimensional PDP for Bike Rentals

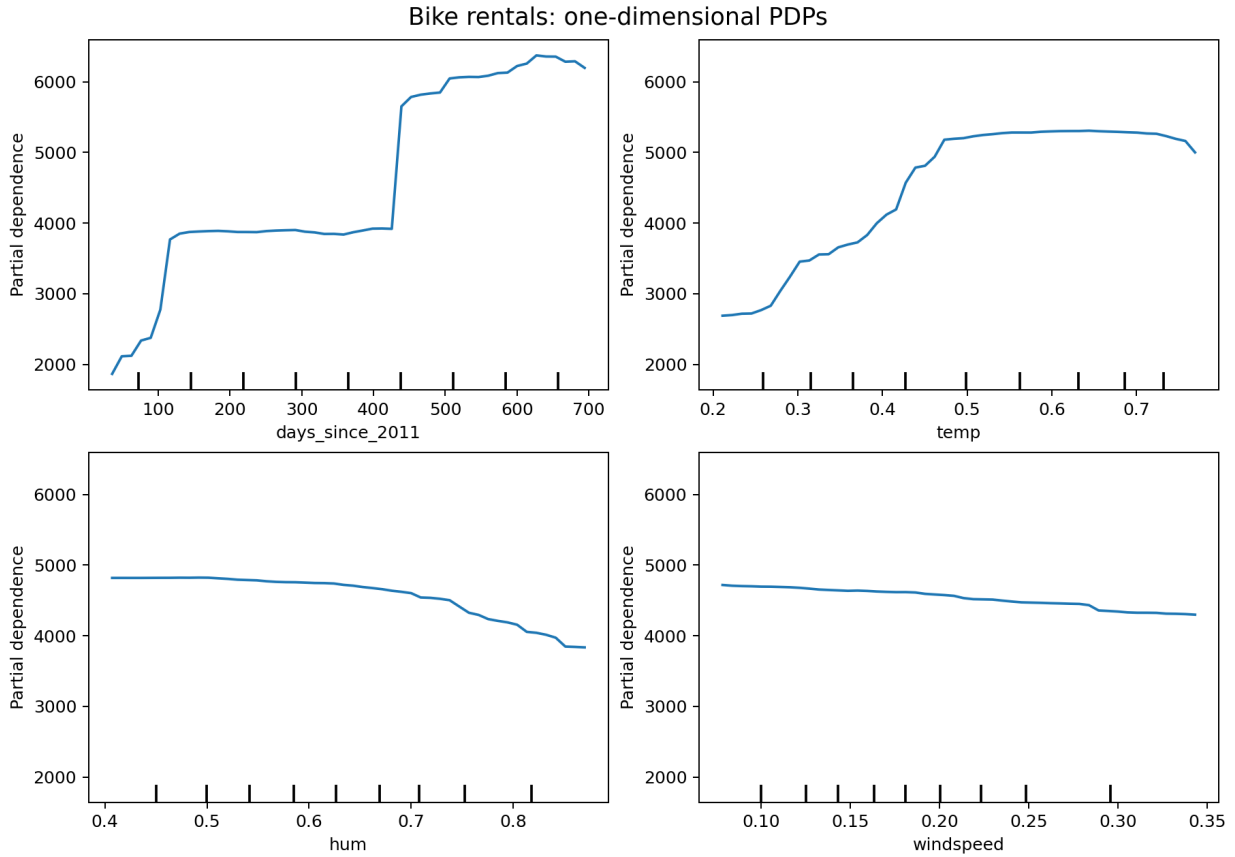


Figure 1: One-dimensional PDPs for days since 2011, temperature, humidity, and wind speed.

Influence of days since 2011. The PDP shows a clear increasing tendency: predicted bike rentals are lower at the beginning of the period and higher later. This suggests that the model learned a temporal growth effect, probably combining real demand growth with yearly and seasonal patterns present in the data.

Influence of temperature. Temperature has a strong positive effect. Predicted bike counts rise as temperature increases from cold to moderate/warm values. The increase becomes weaker at the highest temperatures, so the model does not treat heat as unlimitedly beneficial.

Influence of humidity. Humidity has a negative effect. The model predicts more rentals at low or medium humidity and fewer rentals when humidity becomes high. This is reasonable because humid weather is usually less comfortable for cycling.

Influence of wind speed. Wind speed is mostly negative at higher values. The effect is weaker and noisier than temperature or humidity, but the model still learns that stronger wind tends to reduce expected rentals.

4 Bidimensional PDP for Humidity and Temperature

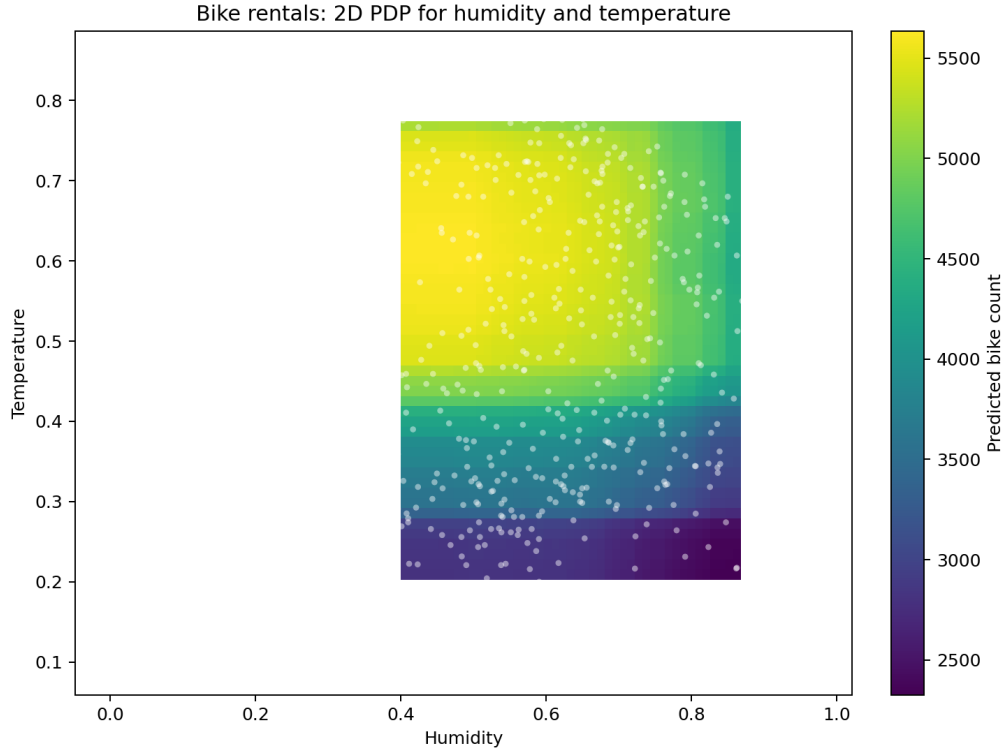


Figure 2: Bidimensional PDP for humidity and temperature. White points show the sampled input distribution.

The 2D PDP confirms that temperature and humidity interact in the bike rental model. The highest predicted rental counts appear in warm conditions with low to medium humidity. The lowest predictions appear in cold conditions and also in areas with high humidity. Temperature is the dominant positive driver, but high humidity reduces the expected demand even when temperature is favorable.

The point overlay shows where observed combinations of humidity and temperature are concentrated. This is important because PDP regions with very few observed points should be interpreted cautiously: they represent model extrapolation more than strongly supported data behavior.

5 PDP for House Price

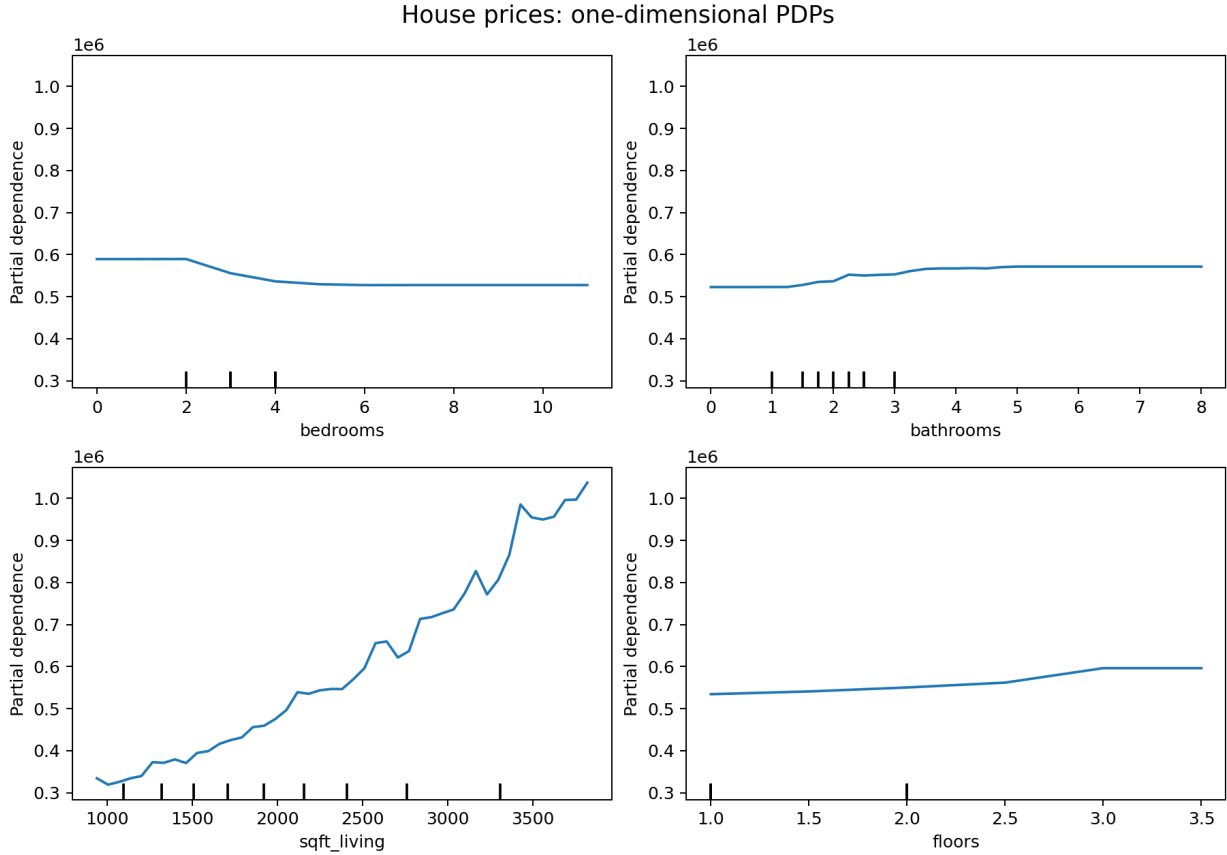


Figure 3: One-dimensional PDPs for bedrooms, bathrooms, living area, and floors.

Influence of bedrooms. Bedrooms do not have a simple strong positive relationship with price. The PDP is comparatively flat and slightly irregular, which means bedroom count alone is not the main driver once bathrooms, living area, lot size, floors, and year built are also included. Very high bedroom counts should be interpreted carefully because they are uncommon.

Influence of bathrooms. Bathrooms have a positive effect. Houses with more bathrooms are predicted to be more expensive, especially when moving from small values to medium/high values. This likely captures both comfort and the fact that larger houses often include more bathrooms.

Influence of sqft_living. Living area is the strongest and clearest predictor among the requested variables. The PDP rises as `sqft_living` increases, meaning the random forest predicts higher prices for larger houses. The increase is steep over common living-area values and then becomes less stable at the largest values, where there are fewer observations.

Influence of floors. Floors have a weaker effect than living area and bathrooms. The model gives somewhat higher predictions for houses with more floors, but the relationship is not as strong or as smooth. This suggests that floor count is useful mainly together with other structural variables.

6 Practical Implications and Limitations

The PDPs are useful because they turn a random forest into readable marginal effects. For bike rentals, the plots can support operational planning: more bikes and rebalancing effort are expected

to be needed on warm, relatively dry days and in the later period of the service. For house prices, the PDPs show that living area and bathrooms are stronger price drivers than bedroom count alone.

There are also limitations. PDPs average over the full data distribution, so they can hide individual-level differences and interactions. The two-dimensional PDP partly solves this for humidity and temperature, but regions with few observed points should still be interpreted carefully. Finally, these models are predictive and explanatory, not causal: the plots show what the fitted random forest learned, not guaranteed causal effects in the real world.

7 Conclusion

The PDP analysis gives interpretable summaries of the random forest models. For bike rentals, time and temperature increase predicted demand, while humidity and wind speed reduce it. The 2D PDP shows that warm and relatively dry weather gives the highest predicted rental counts. For house prices, living area is the dominant positive factor, bathrooms also increase predicted prices, while bedrooms and floors have weaker and less direct effects.